

Reiner Ethan Umila

(647) 550-4317 • umilar@mcmaster.ca • linkedin.com/in/reinerethan-umila/ • github.com/reiner-u • reinerumila.dev

PROFESSIONAL SUMMARY

Software Engineering student at McMaster University currently completing an 8-month Software Quality Assurance co-op with the Government of Ontario, performing manual and automated QA testing on OSAP's student- and administrator-facing web applications, surfacing defects through regression testing following the platform's migration to Azure, overhauling long-neglected test documentation, and building a self-directed Python Playwright script that automates end-to-end test account creation. Hands-on experience also spans full-stack development, data engineering, and embedded systems: designed and deployed live a full-stack Notion-style productivity app (Next.js, TypeScript, Supabase) with a dynamic-schema database and a layered Row Level Security and GRANT permission model, and a separate full-stack weather application with a FastAPI Python back-end, pandas ETL pipeline, Pydantic validation models, SQLite caching, and a fully styled React front-end. Additional experience in algorithmic Python development with NumPy, Streamlit dashboard design, REST API integration, SQL, object-oriented programming, UML design, Selenium web testing coursework, AI-assisted development workflows using Claude, and technical documentation. Hardware background includes ESP32 firmware in C++, custom PCB design in Autodesk Fusion, I²C/SPI/MQTT integration, and Node-RED IoT dashboards from a formal capstone project.

EDUCATION

Bachelor of Technology, Software Engineering Technology (Co-op)

McMaster University

Expected December 2027

Ontario College Advanced Diploma, Computer Engineering Technology

Seneca Polytechnic

2024

- 3.8 GPA, Honours

PROJECTS

Noctus (Notion-Style Productivity App) | Next.js, TypeScript, Tailwind CSS, shadcn/ui, Supabase, Vercel

Live Personal Project

2026 – Present

- Designed, built, and deployed a full-stack, Notion-style productivity app live on Vercel: a dynamic-schema Postgres database (via Supabase) supporting five user-defined column types (text, number, date, boolean, select), with full CRUD across boards, properties, and entries, and type enforcement handled at the database layer rather than the application layer.
- Implemented a two-layer permission model combining Row Level Security policies, some tracing data ownership through two table joins, with explicit Postgres GRANT statements as a separate, independently enforced access check, and built the dynamic table interface on TanStack Table with type-aware column logic driven entirely by each board's user-defined schema.
- Diagnosed a subtle React reconciliation bug where inline render functions changed identity on every data update, remounting table cells and silently discarding in-progress edits; fixed it with a ref and a stable useCallback so the function identity never changes.

Weather Application (Full-Stack) | Python, FastAPI, React, JavaScript, pandas, Pydantic, SQLite, HTML/CSS

Live Personal Project

December 2025 – Present

- Designed, built, and deployed live (Netlify, Render) a complete client/server weather application (v1.0): a FastAPI Python back-end with two RESTful API endpoints, CORS middleware, structured exception handling, and Pydantic data models (CurrentWeather, HourlyWeather, WeatherResponse) enforcing validated response contracts at the API boundary.
- Architected two back-end data retrieval paths: a city-name endpoint geocoding user input via the Nominatim API before querying Open-Meteo across 8 meteorological variables, and a coordinates endpoint accepting raw browser geolocation values with reverse geocoding at town/borough granularity.
- Applied pandas to run an ETL-style data pipeline on raw hourly forecast data: constructing timezone-aware datetime indices from UTC timestamps via tz_convert(), filtering a 168-hour raw dataset to a 24-hour rolling forward window, rounding and type-casting all fields, and serialising the result for front-end consumption; implemented SQLite-backed caching with 1-hour expiry and retry logic to reduce redundant API calls.
- Built a fully styled React front-end with a CSS custom property design system (automatic light and dark mode), card-based hourly forecasts, a hand-rolled SVG precipitation chart, dual-mode location input (browser geolocation plus city search), and async state management via React hooks.

BlockBlast! Puzzle Solver | Python, NumPy, Streamlit

Live Personal Project

February 2026 – Present

- Designed a modular, object-oriented codebase across four files with clean separation of concerns: solver orchestration, board mechanics, piece catalog, and GUI layer, each independently testable and replaceable.
- Built a recursive, permutation-aware search algorithm exploring every ordering of the three available pieces, since placement order affects which lines clear and what space opens up next, applying a depth-first search at each ordering across an 8x8 NumPy board representation to find the highest-scoring complete placement sequence.
- Implemented piece recognition that crops user input to its bounding box and validates it against a catalogue of named NumPy shapes, and built an interactive Streamlit GUI with a clickable board editor and colour-coded, step-by-step solution playback, deployed live on Streamlit Community Cloud.
- Developed a test suite validating solver output against known board states and applied a continuous test-and-refine cycle to surface and fix edge case failures at each development stage.

Human Benchmark Game Device (Capstone) | C++, ESP32-S3, Node-RED, Autodesk Fusion, PlatformIO

Academic Project, Seneca Polytechnic

November 2024

- Collaborated with a partner across the full project lifecycle within a 2.5-month timeline: requirements, system design, C++ firmware on the ESP32-S3, two custom PCBs designed in Autodesk Fusion, hardware integration, testing, and formal delivery to an evaluation panel.
- Diagnosed and resolved multiple hardware integration issues by tracing root causes through datasheet analysis and targeted testing, resolving I²C, SPI, and GPIO conflicts; integrated a Node-RED IoT dashboard for real-time session performance tracking and data logging via MQTT.

WORK EXPERIENCE

Software Quality Assurance Intern (Co-op)

May – December 2026

Ontario Ministry of Public and Business Service Delivery and Procurement, Toronto, ON

- Perform manual and automated QA testing on OSAP's Student-Facing (SFA) and Administrator-Facing (AFA) web applications for the Government of Ontario, covering Full-Time, Part-Time, Microcredential, and OLSG/OLSG-ME financial aid application types across pre-production builds, logging and retesting defects with developers in HP's ALM.
- Took the initiative to build a Python Playwright script that automates end-to-end student profile creation across two QA environments, generating checksum-valid synthetic SINS and logging each new account's OAN from the confirmation page, alongside direct exposure to the team's legacy UFT/VBScript automation tooling.
- Conducted regression testing following the platform's migration from on-premise infrastructure to Azure, surfacing genuine, non-cosmetic defects in older academic-year application logic that had not been re-validated after the move.
- Rewrote and streamlined test case documentation left unrevised for up to five years, correcting unclear or inaccurate pre-conditions and execution steps to improve usability for future co-op cohorts.

Warehouse Attendant

2022 – 2026

University of Toronto Press, Toronto, ON

- Identified the root cause of a persistent inventory backlog in the warehouse management system (AS400), developed a solution independently, and executed both the physical reorganisation and system updates, reducing the required cleanup cycle from weekly to monthly.
- Regularly called upon by colleagues to diagnose and resolve technical issues with Windows workstations, printers, and AS400 warehouse software, drawing on self-acquired knowledge to keep operations running.

ADDITIONAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, C, HTML/CSS, SQL

Frameworks: FastAPI, React, Next.js, Pydantic, Streamlit, pandas, NumPy, Supabase, Tailwind CSS, shadcn/ui, TanStack Table

Testing: Playwright (Python), Selenium (coursework), UFT/VBScript (legacy exposure), HP ALM, regression, functional, and system testing, UML diagrams

Networking & IT: DNS, HTTP/HTTPS, TCP/IP, Linux (Raspberry Pi), Kali Linux, Security Onion, Wireshark, Cisco Packet Tracer, Active Directory

Concepts: OOP, REST APIs, full SDLC, Agile, Waterfall, Iterative, client/server architecture, data modelling, relational database design, ETL/ELT pipelines, QA methodology, STLC, OAuth, Row Level Security

Tools: Git, GitHub Actions, VS Code, PlatformIO, Node-RED, Autodesk Fusion, Vercel, Netlify, Render, Claude (AI-assisted development), Microsoft Office, Teams, SharePoint, AS400